

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	01/10/2026	Duration:	60 minutes

Code of Conduct

I Divya Dharshini V S (192511308) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

AI-Based Network Intrusion Detection and Threat Monitoring System

1. Understanding the Problem Statement

1.1 Explain the problem in your own words.

Nowadays, it has become common for organizations to utilize computer networks to store and share vital information. Thus, the protection of the network from unauthorized intrusion is of paramount importance. In this project, I am proposing an AI-Based Network Intrusion Detection and Threat Monitoring System to detect suspicious activities happening over the network.

- Various cyber-attacks can lead to damage important data, systems, servers and services.
- Attackers may try to penetrate the network illegally or extract vital information from it.
- Some of the common attacks are port scanning, malware traffic, unauthorized access, etc.
- These days, due to the huge volume of traffic flowing in a network, monitoring traffic manually has become really challenging.

- Additionally, the traditional intrusion detection systems mainly rely on the set of rules and patterns for attack detection.
- Thus, detecting new patterns of attack and modified attacks is not really efficient with the current systems.

1.2 Identify the objectives of the proposed system.

The main aim of my project is to build a system that can monitor the traffic over a network and detect potential threats using AI and ML. I intend to make it easy for the administrator in charge of a network to monitor and detect any threats at an early stage.

- Monitor the incoming and outgoing traffic of the network under surveillance constantly.
- Detect any suspicious activities and potential intrusion attempts.
- Scrutinize the packets in a network and detect irregular traffic patterns.
- Leverage AI and ML in order to detect anomalous network behaviors.
- Provide alert mechanisms to the administrator in case of any suspicious activities.
- Document logs of the detected threats for future analysis and report generation purposes.

1.3 State the scope of the problem.

The scope of my project entails building a system that monitors the network activities and detects potential security threats. The project will involve the inspection of the traffic flowing in and out of the network under surveillance.

- Network Monitoring: Monitoring the traffic flowing in and out of a network under surveillance while observing the connection behaviors of the attached devices.
- Traffic Inspection: Inspecting the packets, protocols, connections, and traffic patterns in order to detect any irregularities.
- Abnormal Behavior Detection: Detecting activities and traffic behaviors that deviate from the normal traffic patterns using machine learning.
- Intrusion Detection: Detecting potential intrusion attempts, unauthorized access, and denial-of-service attacks among other security problems.
- Threat Classification: Classifying the detected threats according to their severity levels.
- Alerts: Raising alerts to the system administrator in case of any irregular activities or traffic patterns detected.

2. Identify key stakeholders and challenges

2.1 Identify all the stakeholders involved in the system.

In this system, there are several stake holders. Some will directly benefit from using the system while others will implement and manage the system.

- Security Administrator: The primary user of the alerts in this system. They monitor the network looking for anomalous events to investigate.
- Network Administrator: This stake holder manages the network, devices, and traffic flowing in the network.
- System administrator: They manage the servers and systems that provide the systems resources.
- Cybersecurity team: The group will investigate the suspicious activities identified by the system and respond to major security incidents.
- Employees/Users: The organization's employees will use their networks in their daily operations. This makes their computers vulnerable to attacks.
- Database Administrator: This stake holder will protect and manage the databases that hold the network traffic.

2.2 Explain the roles and responsibilities of each stakeholder.

- Security Administrator: Monitor the alerts produced by the system, Investigate suspicious activities reported by the system, Conduct detailed analysis of possible attacks on the network.
- Network Administrator: Maintains the network devices, Monitors the general traffic in the network, Assists the security team to investigate traffic-related problems, Supports the security team in analyzing the results identified during an investigation
- System Administrator: Manages the systems and servers that host the application, Monitors if all systems are operating as intended, Ensures systems have the correct configurations.
- Cybersecurity team: Investigate the threats and attacks reported by the system, Determine the nature of reported incidents and threats, Assist in responding to major security incidents or attacks.
- Database Administrator: Ensure the database is maintained and managed effectively, Provide guidance on how to protect the data stored in the database, Set up and maintain the databases and systems used to store the network traffic, Recover deleted or corrupted information from the systems and databases

2.3 Discuss the major challenges and issues faced by the stakeholders.

- Large network traffic – Organization's network traffic can be large. Processing this amount of traffic requires an advanced system.
- False alarms – The system may trigger false alarms which will increase the traffic administrators have to investigate.
- New attack methods – Cyber attackers may develop new attack methods which may pose a challenge to the system.
- Detection accuracy – The system should be able to distinguish between normal traffic and anomalous traffic to avoid causing harm.
- Alert timely – The system should generate alerts in a timely manner so that administrators are able to respond to security problems.
- Alert interpretation – Administrators may not be able to interpret the alerts and may take inappropriate action against security threats.
- System protection – The system should be designed in a way that it cannot be compromised by malicious users.

3. Analyze the Current Approach

3.1 Describe the existing system or current process.

In the existing system, the network security is being ensured using various tools like firewall, IDS, antivirus, and human monitoring to detect and prevent the common attacks.

- Firewall: It monitors the traffic in the network and denies the connection if it detects any unwanted or illegal packet.
- IDS: It monitors the network to detect any suspicious activity or traffic that matches the predefined attack signature.
- Antivirus: It mainly protects the computer and server from malware infections.
- Manual Monitoring: This involves continuous tracking, inspection, and detection of network traffic anomalies by the administrator.
- Security Logs: These are the set of all relevant network traffic and security-related activities that can aid the investigation in case of an attack.
- Administrator Action: Once an attack is detected by the automatic tools, the administrator has to analyze whether it is a true or false attack and initiate response accordingly.

3.2 Identify its strengths and limitations.

Strengths:

- A firewall offers fundamental protection to the system connected to it.
- Firewall blocks unwanted or illegal packets to ensure secure network communication.
- IDS helps detect attacks that are known.
- Antivirus helps detect and protects servers from common malware infection.
- Security logs are useful when an attack is in progress or has occurred to help in proper investigation.

Limitations:

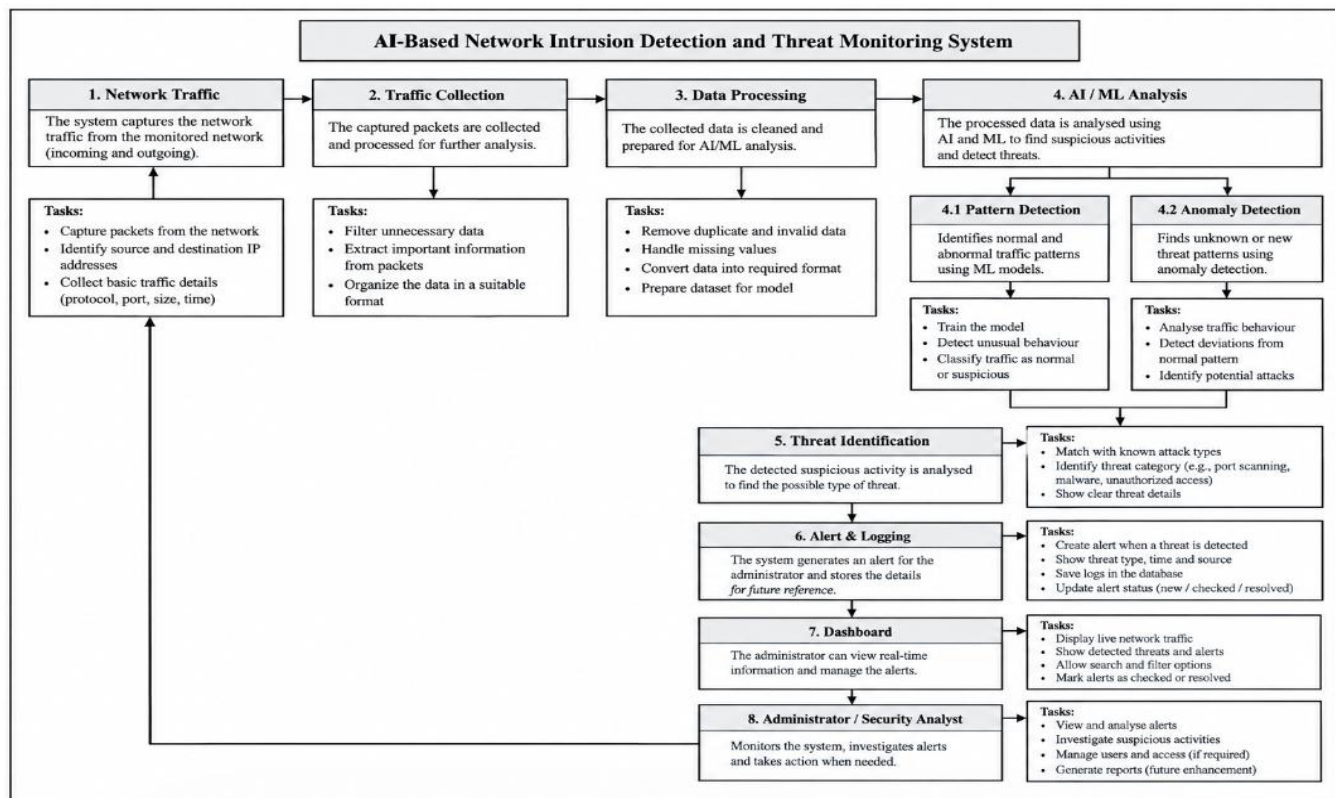
- The existing approach depends on detecting known attacks.
- The manual monitoring and analysis of alerts may be cumbersome, especially with large networks.
- Signatures of false attacks cannot be easily distinguished from those of true attacks.
- Security logs need proper and timely analysis to detect any anomalies.
- The approach can fail to understand abnormal traffic behavior.
- Constant update of signatures and rules is needed for effective attack detection.

3.3 Analyze the problems or gaps that need to be addressed.

- New attack: In case of new attacks, there are no known signatures to help detect them.
- Manual Monitoring: The manual monitoring of network traffic and analysis of alerts may be overwhelming to the administrator.
- False attacks detection: The approach may fail to distinguish between true and false attacks when there are irregularities.
- Slow attack detection: In case an attack is detected after a long time, the damage might be serious.
- No self-learning technique: Traditional approaches use known attack signatures and do not have the ability to learn new traffic behaviors to detect anomalies.
- Need of AI approach: AI and ML techniques can help monitor and analyze the network behaviors to detect anomalies and help offer the needed protection.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.



4.2 Describe the major functional modules of the proposed system.

Traffic Collection is the collection of data about the network traffic going in and out of the network. Data Processing is the process of pre-processing and preparing the data for use. AI/ML refers to the utilization of AI and Machine Learning algorithms to detect irregularities and patterns in the data. Threat Classification is a step of classifying the type of threat detected. Alert is sending out notifications to network administrators and logging the detected incidents.

4.3 Identify the functional and non-functional requirements.

Functional Requirements

- The system should be able to collect network traffic data.
- The system should analyze the traffic data.
- The system should detect any irregular traffic activities.
- The system should send out alerts upon detecting any irregular traffic.
- The system should be accessible only to the appropriate network administrators.

Non-Functional Requirements

- Performance – the system should have high-performance modules to process and analyze.
- Security – network traffic data should be secured and encrypted within the system.
- Reliability – the system should be reliable and have no or minimal errors.
- Usability – the system should have a user-friendly dashboard to display the detected irregularities.
- Accuracy – the system should have high accuracy in detecting network intrusions.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

- Modularity: The system is modular. It is easy to maintain and manage modules.
- Scalability: The system is scalable. It is easier to manage modules based on the network's size.
- Maintainability: The system has a modular structure. It makes it easy to update a module.
- Reliability: The system undergoes proper testing and has error-handling mechanisms.
- Usability: The system has an intuitive dashboard to offer a short learning curve.
- Security: The system has secure logins and access controls to protect data.
- Flexibility: The system uses AI/ML models that can be updated to detect new network attacks.

5. Justify the Chosen Methodology/Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

When approaching this project, I would select the Agile methodology that is aimed at breaking down the software development process into smaller segments of work. I will be able to develop and test each module of the system separately to build a comprehensive network security application.

- The project will be divided into several small sprints.
- Each sprint will tackle a particular module or feature of the software.
- The first sprints will involve collecting and processing the data.
- I will then proceed to developing and testing the AI/ML detection module.
- Each completed segment will allow me to make adjustments before proceeding to the next.
- Throughout the process, I will be able to gather feedback and incorporate it into my work.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

- The new security features and requirements can be added at any point during development.
- The same is true for the AI/ML model that I plan to use for the system.
- I will be able to conduct regular tests and ensure that no errors have been introduced.
- The process allows for continuous improvements and adjustments.
- Administrators can provide feedback regarding the dashboard and alerts at any stage of the process.
- Finally, making small-scale adjustments is easier and less time consuming than overhauling the system at the end of development.

5.3 Explain how it supports successful project development and maintenance.

- Agile is ideal for development and maintenance as it allows for continuous improvement.
- I will be able to add new features and make adjustments to the codebase.
- This is an important consideration for my project as the security space will continue to evolve.
- I can utilize the sprints to further develop and refine each module of the system.
- New security features and tools can easily be integrated into the application.
- The AI/ML component will be adjustable to accommodate new data and requirements.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

The designed system will offer certain benefits, including improving network security and reducing the amount of time that the administrator spends on monitoring. The system will be able to detect suspicious activities in the network and provide informative insights.

- The system will be able to detect suspicious activities in the network.
- This system will provide alerts in case of detecting any threats.
- The designed system will reduce the amount of manual work of the administrator.
- The system will store all the detected threats' data for the administrator's future reference.
- The system will help the administrator identify any potential security issues in the early stages.
- The administrator will be able to refer to the data about the detected threats to get insight into the network traffic.

6.2 Explain how the solution addresses the identified challenges.

There are several challenges that the system might face. The system will have to process a lot of data concerning the network traffic, which will put a lot of pressure on the AI model. The AI model will need sufficient data to offer reliable results, and the administrator will have to double-check the detected threats before reacting to them.

- The system will have to process a lot of data about the network's traffic.
- The AI model will need sufficient data to offer reliable results.
- The system might generate false positive results.
- The administrator will have to double-check the detected threats before reacting to them.
- Setting up the system properly to make it compatible with the network will be challenging.

6.3 Discuss potential risks, implementation challenges, and limitations.

- The system will not be able to detect all kinds of attacks.
- There is a possibility that the system will generate false positive results.
- The system will require up-to-date data about threats and attacks.
- The system requires proper maintenance and supervision of the administrator.
- There is a possibility that the system will occupy too much memory and space by the data that it will store.

6.4 Suggest possible future enhancements.

- Connecting the system to the firewall to make it automatically block the detected threats.
- Retraining the AI model with more protective data sets to make it more efficient.
- Adding an email or mobile notification function that will notify the administrator about the detected threats.
- Making the dashboard more informative by adding visual elements such as pictures and charts.
- Enabling the system to respond to the detected threats automatically when they are serious.
- Adding more network security tools to the system design.
- Expanding the attack data set that the system will use to detect network threats and suspicious activities.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25